

The exact local non-Gaussianity of a matter-dominated contraction: $f_{\text{NL}} = -35/16$ and its orientation dependence

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A matter-dominated cosmological contraction ($w = 0$, $\epsilon = 3/2$) generates local primordial non-Gaussianity through the cubic self-interaction of the comoving-gauge curvature perturbation ζ . We report a from-scratch in-in computation of every cubic vertex in Maldacena’s action for this background, validated against the de Sitter and ultra-slow-roll limits before use. The isocles squeezed-limit amplitude is $f_{\text{NL}}^{\text{local}} = -35/16$, confirming the value obtained by Li, Quintin, Wang and Cai (2016) at unit sound speed. We trace the long-standing discrepancy with $-35/8$, originally reported by Cai, Xue, Brandenberger and Zhang (2009), to a uniform factor of two in that work’s amplitude-normalization step; the underlying shape function is correct term by term. We report a new result: the squeezed limit is orientation dependent, $f_{\text{NL}}(\mu) = -35/16 + (15/16)\mu^2$ with μ the cosine between the long and short wavevectors (monopole $-15/8$, quadrupole $15/16$), sourced by the shear of the non-attractor growing mode and analogous to the consistency-relation violation identified by Namjoo, Firouzjahi and Sasaki in ultra-slow-roll inflation. A separate δN calculation on uniform-density slices returns $(5\epsilon - 35)/8 = -55/16$; this is a different physical quantity, not a discrepancy, reconciled by $\zeta_\rho = 2\zeta_c$ at linear order in a non-attractor phase. Survival of this amplitude through an explicit nonsingular bounce is bounded, not resolved: linear transfer across three bounce backgrounds obeys a universal suppression $0 < T_{f_{\text{NL}}} \leq 1/2$, scheme-dependent at the $1.6\times$ level, while the bounce’s own cubic term – found in the loop-quantum-cosmology literature to enhance non-Gaussianity by orders of magnitude – is not computed here. Only a survey-scale bispectrum measurement can currently distinguish $-35/16$ from $-35/8$: both fit DESI DR1 (0.16σ), while SPHEREx reaches $2.6\text{--}3.7\sigma$ between them.

I. SETUP

We work in conformal time $\eta < 0$ with $M_{\text{Pl}} = 1$, unit sound speed, and a matter contraction $a(\eta) = c\eta^2$, for which the Hubble slow-roll parameter is fixed, $\epsilon \equiv 1 - \mathcal{H}'/\mathcal{H}^2 = 3/2$, with $\mathcal{H} \equiv a'/a < 0$. The quadratic action for the comoving-gauge curvature perturbation ζ ($h_{ij} = a^2 e^{2\zeta} \delta_{ij}$, uniform-field gauge $\delta\phi = 0$) is $S_2 = \int d\eta d^3x a^2 \epsilon [\zeta'^2 - (\partial\zeta)^2]$, with Bunch–Davies mode functions fixed by the Wronskian normalization (not assumed). On super-Hubble scales the physical mode is the growing, non-attractor solution $\zeta' = -3\zeta/\eta$ (equivalently $\dot{\zeta} = -\frac{3}{2}H\zeta$), the analogue of the growing mode used by Cai *et al.* [1].

We define the local bispectrum amplitude in the isocles squeezed configuration $k_1 \ll k_2 = k_3 \equiv k$ exactly as in Refs. [1, 2]:

$$f_{\text{NL}}^{\text{local}} \equiv \lim_{k_1 \rightarrow 0} \frac{10}{3} \frac{\mathcal{A}(k_1, k, k)}{2k^3}, \quad (1)$$

where $\langle \zeta_{\mathbf{k}_1} \zeta_{\mathbf{k}_2} \zeta_{\mathbf{k}_3} \rangle = (2\pi)^7 \delta^3(\sum \mathbf{k}_i) \mathcal{P}_\zeta^2 \mathcal{A} / \prod k_i^3$, equivalent to the local template $\zeta = \zeta_g + \frac{3}{5} f_{\text{NL}} \zeta_g^2$. This limit is independent of the end-of-contraction time η_B at leading order ($\mathcal{O}(k^2 \eta_B^2)$ corrections) but, as we show below, is not independent of the orientation of the short pair relative to the long mode.

We use the cubic action derived by Maldacena [3] in comoving gauge for a canonical field, retaining every term at $\eta_{\text{sr}} = 0$: two bulk pieces proportional to ϵ^2 , one boundary/field-redefinition term, and the two ϵ^3 pieces that involve $\tilde{\chi} \equiv \partial^{-2}\zeta'$. All four bulk vertices and the field redefinition are evaluated by first-order in-in perturbation theory, $\langle \zeta^3(\eta_*) \rangle = -2 \text{Im} \int^{\eta_*} d\eta \langle \zeta^3(\eta_*) L_{\text{int}}(\eta) \rangle$, with every Wick contraction summed explicitly (no symmetry factor inserted by hand) and every time integral performed exactly before taking $|k\eta_B| \rightarrow 0$.

II. VALIDATION

Before applying this pipeline to the matter contraction we reproduce two independent literature results with the same code: (i) the de Sitter, constant- ϵ bispectrum of Ref. [3], matched term by term; and (ii) the ultra-slow-roll result of Namjoo, Firouzjahi and Sasaki [4], $f_{\text{NL}} = 5/2$, which depends on the same field-redefinition structure that we use below. Both match exactly.

III. RESULT

Table I gives the squeezed-limit contribution of each vertex, from scratch, at $\epsilon = 3/2$. The total reproduces the isocles squeezed value

$$f_{\text{NL}}^{\text{local}} \Big|_{k_1 \ll k_2 = k_3} = -\frac{35}{16} \quad (2)$$

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TABLE I. Squeezed-limit contribution of each cubic vertex at $\epsilon = 3/2$, from a from-scratch in-in computation. μ -dependence is shown where nonzero.

vertex	$f^{\text{sq}}(\mu)$
field redefinition (local + non-local part)	$-\frac{25}{16} - \frac{15}{16}\mu^2$
$\zeta\zeta'^2$ ($\epsilon^2 - \epsilon^3/2$)	$-\frac{5}{32}$
$\zeta(\partial\zeta)^2$	0
$\zeta'\partial\zeta\cdot\partial\tilde{\chi}$	$\frac{15}{8}\mu^2$
$\zeta(\partial_i\partial_j\tilde{\chi})^2$	$-\frac{15}{32}$
total	$-\frac{35}{16} + \frac{15}{16}\mu^2$

and, at general orientation $\mu \equiv \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}$,

$$f_{\text{NL}}^{\text{sq}}(\mu) = -\frac{35}{16} + \frac{15}{16}\mu^2, \quad (3)$$

i.e. a monopole $-15/8$ and a quadrupole $15/16$ under angular averaging. The equilateral and folded values are $-255/128$ and $-9/8$, matching Ref. [2]’s $c_s = 1$ limit exactly.

Origin of the discrepancy with Cai et al. [1]. Reading their printed shape function (their Eq. 37) as the sum of three distinct $(5, 2, 2)$ -type monomials, it is identical to our vertex sum, and to Li et al.’s Eq. 4.19 at $c_s = 1$ (difference zero, checked symbolically). Cai et al.’s three quoted amplitudes – local $-35/8$, equilateral $-255/64$, folded $-9/4$ – are each exactly twice the values in this Letter, in all three configurations. The origin is a uniform factor of two carried into their amplitude-normalization step, not an error in any vertex, commutator, or Wick contraction. Li et al. [2] inherit Cai et al.’s correct polynomial and are therefore not an independent check; Quintin et al. [5] cite $-35/16$ without an independent computation. The present result is, to our knowledge, the first computation of this amplitude that does not transcribe a per-vertex expression from either.

Orientation dependence and its origin. Equation (3) arises because the growing, non-attractor mode is not locally an FRW patch as $k_1 \rightarrow 0$: its extrinsic-curvature perturbation carries both a trace, $\delta K \rightarrow \epsilon \dot{\zeta}$, and a shear of the same order, $\sigma^i_j = (\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij}) \epsilon \dot{\zeta}$, sourced by the comoving-gauge shift $\partial \tilde{\chi}_L \propto \hat{\mathbf{k}}_1 \zeta'_L / k_1$. The μ^2 term comes entirely from the two vertices in which the long mode enters through $\partial \tilde{\chi}$. This is the matter-contraction analogue of the consistency-relation violation Namjoo, Firouzjahi and Sasaki [4] found in ultra-slow-roll inflation, and is consistent with the general non-attractor clock argument of Chen, Namjoo and Wang [6]: a squeezed-limit amplitude that survives the $k_1 \rightarrow 0$ limit is the generic signature of a non-attractor background, not an artifact of this one.

Cross-check by a different route. A separate-universe (δN) calculation on the same background but evaluated on *uniform-density* slices gives, for general ϵ , $f_{\text{NL}}^\rho =$

$(5\epsilon - 7) \cdot 5/8$ reduced at $\epsilon = 3/2$ to $(5\epsilon - 35)/8 = -55/16$, while the comoving-slice δN (an isotropic, shear-free approximation) gives $f_{\text{NL}}^\epsilon = -5$ for all ϵ . Neither equals the in-in result above, and this is expected rather than alarming: at linear order $\zeta_\rho = 2\zeta_\epsilon$ in a non-attractor phase, so uniform-density and comoving slices differ already at $\mathcal{O}(1)$; and the isotropic separate-universe approximation discards the shear that produces Eq. (3) by construction, keeping only the trace. The δN values are correct for their own variables and are not forced to agree with the in-in result; they instead confirm that the gap between $-35/16$ and Cai et al.’s $-35/8$ is a clean factor of two rather than a definitional mismatch, and they leave a single well-posed open calculation – an anisotropic (Bianchi-I) separate-universe treatment of the long mode’s shear response – as the remaining independent check of the monopole and quadrupole in Eq. (3).

IV. TRANSMISSION THROUGH THE BOUNCE: STATED HONESTLY

Equation (3) is a statement about the contraction phase alone. Whether it survives into an observable post-bounce spectrum depends on physics this Letter does not compute in full, and we state the current evidence precisely. Linear mode-mixing across three explicit nonsingular bounce backgrounds – an LQC effective quasi-dust bounce, an analytic non-LQC bounce, and the Quintin et al. [5]-type parametrization – gives a transmission coefficient $T_{f_{\text{NL}}} = [1 - \rho(\eta_h)]/2$ for the growing-mode-dominated adiabatic vacuum, obeying a universal bound $0 < T_{f_{\text{NL}}} \leq 1/2$: linear transfer can only suppress the amplitude, by scheme- and background-dependent factors in the range 0.165 – 0.409 (a $1.64\times$ spread between two mode-function conventions on the same background). This supersedes an earlier, narrower statement about the transfer of the sub-dominant constant branch alone. What this Letter does not compute is the bounce’s own cubic self-interaction term, which the loop-quantum-cosmology literature finds can *enhance* non-Gaussianity by orders of magnitude through gravitational self-interactions in the third-order Hamiltonian, even in the absence of a potential [7]. Any assumption that the bounce transmits f_{NL} with a coefficient near unity is therefore not supported by either calculation reported here.

V. OBSERVATIONAL REACH

Table II compares the two candidate amplitudes against current and near-future large-scale-structure constraints, using the scale-dependent-bias signature of local non-Gaussianity [8]. DESI DR1 cannot currently distinguish $-35/16$ from $-35/8$; SPHEREx is the first instrument projected to.

SPHEREx sensitivities are quoted from Ref. [9]

TABLE II. Discriminating power of current and near-future bispectrum/scale-dependent-bias constraints on $f_{\text{NL}}^{\text{local}} = -35/16$ vs. $-35/8$.

survey	$\sigma_{f_{\text{NL}}}$	tension, $-35/16$	separation
DESI DR1	–	0.16σ	$\lesssim 0.2\sigma$
SPHEREx, bispectrum	0.7	–	$2.6\text{--}3.1\sigma$
SPHEREx, target ($P+B$)	0.5	–	$3.7\text{--}4.4\sigma$

(§ 2311.13082, abstract) and Ref. [10]; DESI DR1 comparison uses the public headline scale-dependent-bias constraint. Full construction of both channels, plus a Press–Schechter-level primordial-black-hole comparison against Ref. [11] and a PTA free-spectrum consistency check against the induced-gravitational-wave slope of Ref. [12], are given in Ref. [13].

VI. SUMMARY

A from-scratch in-in computation, validated against two independent literature limits, gives $f_{\text{NL}}^{\text{local}} = -35/16$ for a canonical matter-dominated contraction, with a new orientation-dependent squeezed limit $-35/16 + (15/16)\mu^2$ sourced by the non-attractor shear. The long-discussed factor-of-two discrepancy with Ref. [1] is traced to a single normalization step, not to the underlying shape function, which is correct. What remains open is not the amplitude at the end of the contraction but its fate through the bounce: the linear transfer is bounded, $T_{f_{\text{NL}}} \leq 1/2$, and the bounce’s intrinsic nonlinear contribution is not yet computed. Only a survey-scale bispectrum measure-

ment, at the sensitivity SPHEREx targets, can currently discriminate the corrected amplitude from the value it replaces.

DATA AND CODE AVAILABILITY

The from-scratch symbolic derivation, its two literature validation checks, the separate-universe cross-check, the bounce-transmission computation, and the multi-channel reach calculation are committed at <https://github.com/Hubify-Projects/bigbounce> under `research/theory_audit/`, `research/cubic.bounce.transmission/`, and `research/track_a3.multichannel/` (commits referenced in Refs. [13–15]). The parent forecast paper and its own code/data are permanently archived at Zenodo, DOI 10.5281/zenodo.21461881 [16].

AI USAGE DISCLOSURE

Symbolic derivations reported here were performed by deterministic sympy scripts written and executed under the author’s direction; every intermediate result is printed by the committed scripts and independently checked against the cited literature limits before use. Claude (Anthropic, Opus-tier) assisted with literature cross-checks, prose drafting, and LaTeX formatting under the author’s direction; all physics content, equations, and numerical results were verified by the author against the committed script output.

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